

Assignment 2 Report
Data Warehousing/Visualization (CPSC-510-3)

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1. Dataset Overview

1.1 Provided Dataset Description

The dataset provided contains of five CSV files as follows:

FactOrders: transactional data including sales, profit, quantity, shipping, order dates, and foreign keys to Customer, Product, Date, and Region dimensions. Its primary key is OrderID.

DimProduct: provides product attributes such as category, subcategory and productName, Its primary key is ProductID.

DimCustomer: Dimension table that provides geographic information about the customer such as Country, city, state. Additionally, it includes the column segmentation to help classify clients into distinct business categories. Its primary key is CustomerID

DimRegion: geographic classification including region, state, and country. Key restrictions described in data model design.

DimDate: calendar dimension used for time based analysis. Primary key Date

1.2 External Dataset Integration (Population)

To enrich the geographic analysis, an external dataset containing population by country was incorporated. Population figures were obtained from official statistical sources: (Statistics Canada, 2021 and United States Census Bureau, 2021).

The dataset was integrated into the Power BI model using the Country field as the common key, allowing the population table to be related with the Region dimension. This integration allows the creation of additional metrics, such as Sales per 1,000 people, which calculates total revenue by population size and allows for a more comparison between countries.

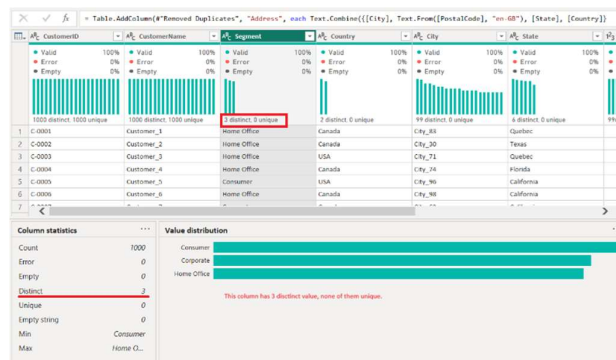
It should be mentioned that population numbers correspond to a single census year (2021). Therefore, the revenue per capita metric represents an approximation, as population changes over time and may not perfectly reflect the exact population during each sales period.

2. Power Query - Data Cleaning & Transformation

All tables were renamed for better clarity. After this step, each table was inspected in power query using the column quality and value distribution (Figure 1) looking for missing values, error or inconsistencies. The analysis confirmed that key columns such as OrderID, CustomerID, ProductID, region and Date fields contained no null or error values. This inspection in value distribution also helped to identify the distinct values of each column.

Additionally, Column Distribution and Column Profile tools were used to examine value ranges and identify potential outliers in numeric fields such as Sales, Profit, and Quantity. No extreme anomalies or invalid values were detected.

Figure 1 Column Quality, statistics and value distribution used in each column



For every column in all the tables the data types were checked and corrected where needed, data types were corrected (dates formatted properly, numeric fields set to decimal, currency columns changed to fixed decimal number).

New columns were created such as Quarter to return quarter values (1–4) for time-based aggregation and Customer Address Creation by merging City, Postal Code, State, and Country and using a comma separator. All transformation steps were documented in Power Query.

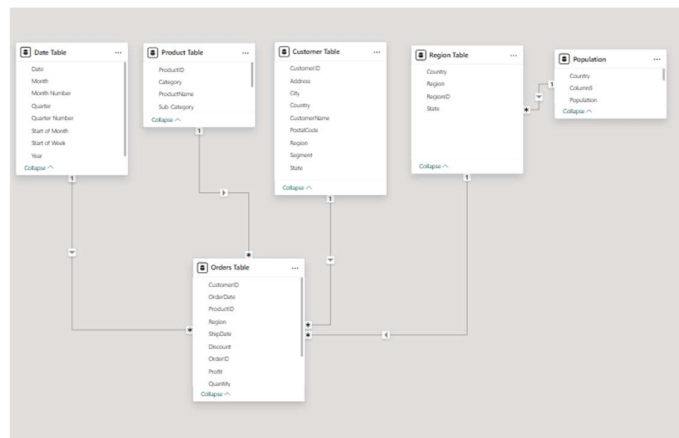
3. Data Model Design

3.1 Star Schema Model

The core data model follows a star schema design with the Orders table as the central fact table connected to the Date, Product, Customer, and Region dimensions. An additional Population table was integrated and linked to the Region table, this creates a small snowflake extension while maintaining the overall star schema structure.

Each dimension is connected directly to the Orders table using one-to-many relationships. Cross-filter direction was set to single to maintain model clarity and performance. The model is seeing in Figure 2.

Figure 2 Model diagram



3.2 Model Validation

A Matrix visual using fields from all tables was created to confirm proper filter propagation across dimensions. Additionally, a validation test was conducted comparing Region from Customer and Region from Region Table. While total sales remained consistent, distribution differences confirmed geographic inconsistencies. Therefore, the Region Table was used as the official geographic dimension.

4. DAX Measures and calculated column

Measures were written using DAX functions as DIVIDE() to safely handle division by zero and DISTINCTCOUNT() for accurate aggregation. Variables (VAR) were used where necessary to improve readability and performance. All measures are dynamic and respond to slicers, ensuring accurate aggregation across the star schema model. The following measures were created in the Measures1 Table:

Total Sales: Sum of all sales in Orders table.

Total Quantity: Aggregates the total number of product units sold by summing the quantity column in the orders table.

Total Profit: Sum of all profit values generated from sales transactions.

Total Orders: Count of distinct order IDs (unique sales transactions).

Total Customers: Count of distinct customers who placed at least one order (from orders).

Average Sales per Client: Total sales divided by the number of distinct customers.

Average Order Value: Total sales divided by total number of orders.

Average Shipping Time: Average number of days between order date and ship date, measuring operational efficiency.

% Orders Within 3 Days: Divides the number of orders shipped within three days by total orders to measure speed performance.

Top Category by Profit: Identifies the product category that generates the highest total profit within the selected filter context.

Sales Growth %: Measures the percentage change in total sales compared to the previous period year, this indicates business growth performance.

Sales % of Total by Product: Represents each product's contribution to total sales as a percentage of overall revenue.

Profit Margin %: Total profit divided by total sales, measuring profitability efficiency per dollar of revenue.

Sales per 1000 people: Divides total sales by the population and multiplies by 1,000 to compare market performance relative to population size.

Average Sale Price (ASP): Total sales divided by total quantity sold, representing the average price per unit.

Average Order Value: Total sales divided by the total number of orders, indicating the average revenue generated per order.

Sales per Day: Total sales divided by the number of days in the selected period, used to normalize sales performance across months with different numbers of days.

Additionally, a calculated column was created to count the difference in days between order date and ship date (Shipping Days = ShipDate - OrderDate). This enables shipping performance analysis.

5. Dashboard Design and Descriptive Analytics

The dashboard was designed following a minimalist approach, using a limited number of visualizations and tables per page. This design choice helps avoid information overload and improves readability, allowing users to quickly interpret the key insights presented in each section.

5.1 Dashboard Pages Description

A homepage was created with multiple buttons to navigate to the multiple dashboards that were created, below is a description of the information available on each page.

Executive Overview: Provides high-level KPIs including Units Sold, Average Sale Price, Average order value, Revenue, and profit Margin. Trend visualizations and country map.

Customer Analysis: Examines customer segmentation, trendline of customers and top 10 Customers per Sales by state.

Product Performance: Evaluates category performance, margin efficiency, and product contribution. This page has views one for Retail Buyer and one for Product Manager.

Logistic Performance: Measures operational performance including shipping time by region and mode.

Data Diagnostic: Contains visuals and tables specifically created to support deeper diagnostic analysis of the key insights identified in the dashboard. This page includes a Decomposition Tree visual to investigate underlying factors behind changes in performance, allowing sales to be broken down by dimensions such as region, category, and time. It also includes regional growth analysis to further examine the drivers behind the observed trends.

User guide: Provides a brief explanation of how to navigate and interact with the dashboard, including the use of filters, drill-down features, tooltips, and navigation buttons to explore the data. Additionally, explaining the KPIS.

5.2 Slicers

Interactive slicers were added to filter by date, quarter, month, country, shipping mode and segment. Additionally, hierarchies were created to allow drill-down functionality. For example, product analysis allows users to drill from Category to Sub-Category, facilitating deeper exploration of sales. This supports interactive data exploration.

5.3 BookMarks

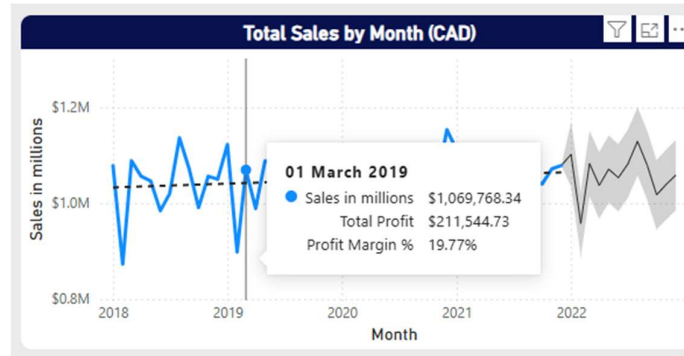
In the Product Performance page, bookmarks were implemented to simulate two user perspectives. The Retail Buyer view focuses on product demand and the Product Manager view highlights profitability and margin performance. This design allows the dashboard to support different decision-making roles. The Main KPIS cards are shown in the top for both users.

5.4 Tooltips

Tooltips were added to the charts to improve readability, as an example Figure 3 shows the monthly sales trend visual to display additional metrics, including Total Profit and Profit

Margin, allowing users to access more detailed performance information when hovering over each data point.

Figure 3 Tooltip Example Used in Charts



5.5 Narrative key insights

A key insights section that summarizes the most important findings from the visuals were added to each page. This can help stakeholders understand trends, patterns and key business implications without needing to analyze each chart individually. The insights were written manually, as the narrative automatically generated by Power BI did not provide meaningful analytical interpretation.

5.6 Scatter Plot – Data Limitation

A scatter plot comparing Average Selling Price (ASP) and product rating was considered. However, this visualization was not possible because rating data was not available in the dataset. Instead, ASP was analyzed in relation to the total quantity sold by product to explore price demand patterns.

5.7 Last Update

An Update Date was included in the home page to show the most recent data refresh. This allows users to quickly verify the timeliness of the information and understand when the data was last updated, improving transparency and reliability in any analysis.

6. Key Insights and Diagnostics.

Monthly Sales Diagnostic

The monthly sales trend analysis showed that February consistently records lower sales compared to other months, indicating a recurring seasonal slowdown (**Error! Reference source not found.**). To ensure that this decline was not simply caused by the shorter number of days in February, a Sales per Day metric was calculated by dividing total monthly sales by the number of days in each month. The results in Figure 5 (added to the data diagnostic page) show that February still presents lower daily sales compared to other months, confirming that the decline is not only explained by the number days in the month.

Figure 4 Total Sales by Month

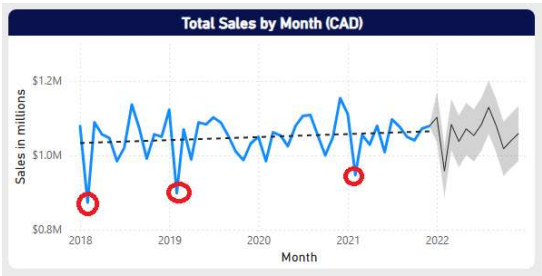


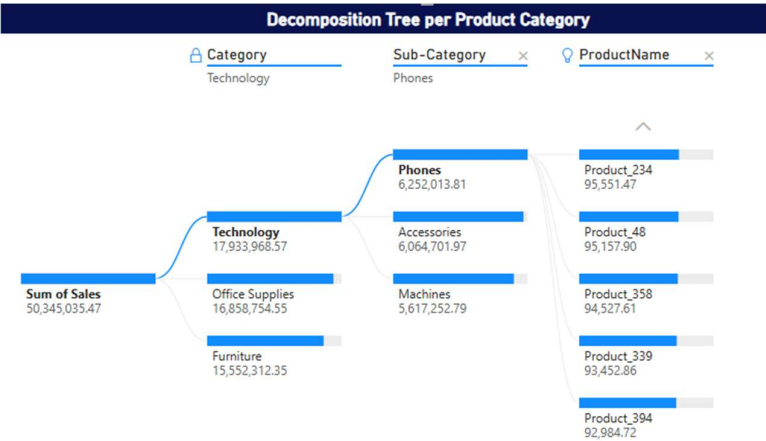
Figure 5 Sales per day for every month

Sales per Day	
Start of Month	Sales per Day
01 January 2018	\$34,801.80
01 February 2018	\$31,180.44
01 March 2018	\$35,119.87
01 April 2018	\$35,207.43
01 May 2018	\$33,770.29
01 June 2018	\$32,813.57
01 July 2018	\$32,917.71
01 August 2018	\$36,667.23
01 September 2018	\$35,726.18
Total	\$34,459.30

To better understand the factors contributing to this decline, a diagnostic analysis was conducted using the Decomposition Tree visual in the data diagnostic page (Figure 6). This tool allowed sales to be decomposed by different dimensions, including product category, sub-category, and region, to identify whether a specific segment was causing the drop in February

sales. Filters were first applied to observe the overall behavior of sales throughout the year and then applied the filter to February month. The analysis showed that the decrease occurs relatively consistently across categories, indicating that no single product category disproportionately contributes to the decline. This suggests that the lower sales observed in February are more likely associated with an overall seasonal pattern rather than a problem related to a specific product group. Because of this, it is recommended that the company consider implementing targeted promotions or marketing initiatives during February to help mitigate the seasonal impact and maintain more stable sales performance.

Figure 6 *Decomposition Tree Analysis*

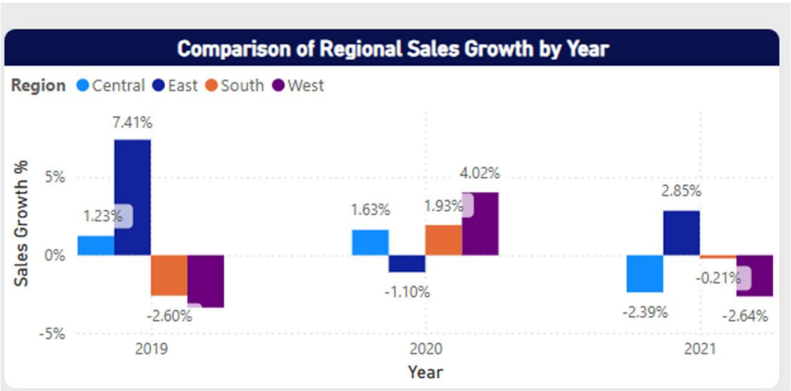


Year Over year Gwoth diagnostic

The year-over-year growth analysis shows that although sales increased in 2020 with a growth rate of 1.60%, the performance declined in 2021, reaching -0.61%. To better understand the drivers of this decreased, sales growth was analyzed by region (Figure 7). The regional breakdown reveals that the decline was mainly driven by negative performance in the Central and West regions, which have negative growhts rates of -2.39% and -2.64% respectively. Interestingly, both regions experienced growth in 2020, suggesting that the decline in 2021 may reflect a slowdown after a period of previous expansion rather than a sudden structural drop in demand. In contrast, the East region show positive growth, indicating that the slowdown was

not uniform across the business but concentrated in specific geographic markets. This diagnostic suggests that the overall decline in 2021 was largely influenced by regional dynamics rather than a company-wide decrease in demand. Based on this finding, it is recommended that the company further investigate market conditions and sales strategies in the Central and West regions to better understand the factors affecting performance and identify opportunities to stabilize growth in those markets.

Figure 7 Comparison of Regional Sales Growth by Year



Sales per 1,000 people insight

Even though the United States generates more total revenue, Canada performs better when we calculate sales by population. Sales per 1,000 people are much higher in Canada \$345 compared to USA %113 (Figure 8), suggesting stronger demand in Canada.

Figure 8 Sales per 1000 people and Revenue per Country.



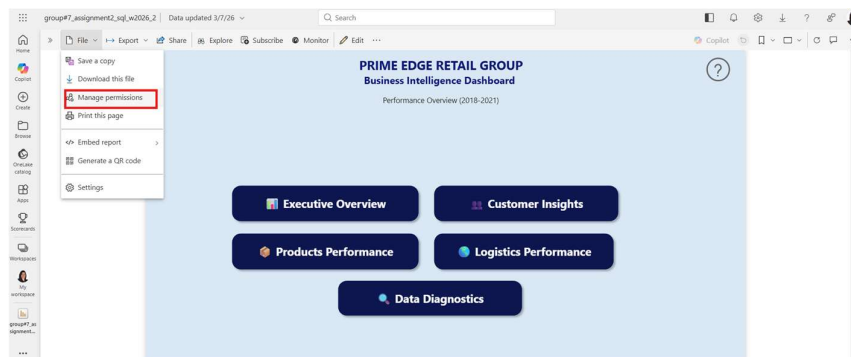
Additional insights are found in the key insights text at the bottom of each page of the report.

7. Interactivity, Publishing & Security

The dashboard incorporates many interactive features to support flexible data exploration. Slicers were implemented to allow users to filter the analysis by date, country, ship mode and segment. Visuals also support cross-filtering, meaning that selecting an element in a chart automatically updates the other visuals on the page. In addition, drill-down functionality and hierarchical fields were used in selected visuals, allowing users to explore data at different levels of detail such as category, sub-category, and product.

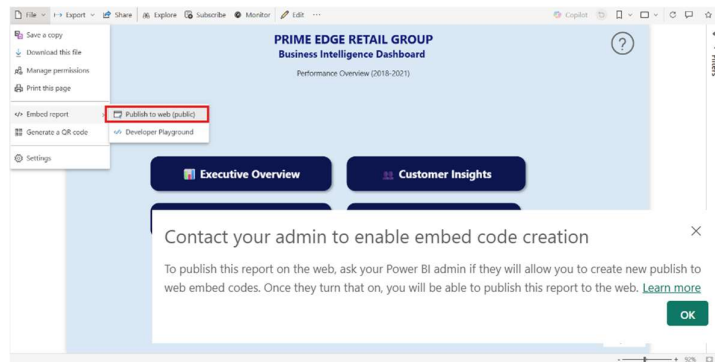
The dashboard was published to the Power BI Service within a personal workspace. In a production environment, the report could be shared with other users through workspace access using Power BI Pro, where permissions can be managed through roles such as Viewer, Contributor, Member, or Admin. Access rights can be configured through the Manage Permissions settings, as shown in Figure 9, allowing administrators to control who can view or modify the report. This process requires a paid Power BI license.

Figure 9 Power bi publication and managing permissions



Power BI also allows reports to be shared publicly through the Publish to Web feature, although this option is not recommended for confidential or sensitive data, in this case the educational licence does not allow this kind of publication, however the Figure 10 shows the option where it can be done.

Figure 10 Publish to web (public)



References

Statistics Canada. (2021). *Census of Population*. Retrieved from <https://www12.statcan.gc.ca/census-recensement/index-eng.cfm>

United States Census Bureau. (2021). *U.S. and World Population Clock*. Retrieved from <https://www.census.gov/popclock/>